

- [17] Yeonkyoung So, Gyuseong Lee, Sungmok Jung, Joonhak Lee, JiA Kang, Sangho Kim, and Jaejin Lee. Thunder-nubench: A benchmark for llms’ sentence-level negation understanding, 2025.
- [18] Christina Steindl, Eva Jonas, Sandra Sittenthaler, Eva Traut-Mattausch, and Jeff Greenberg. Understanding psychological reactance. *Zeitschrift für Psychologie*, 2015.
- [19] Neeraj Varshney, Satyam Raj, Venkatesh Mishra, Agneet Chatterjee, Ritika Sarkar, Amir Saeidi, and Chitta Baral. Investigating and addressing hallucinations of llms in tasks involving negation. *arXiv preprint arXiv:2406.05494*, 2024.
- [20] Tereza Vrabcová, Marek Kadlčík, Petr Sojka, Michal Štefánik, and Michal Spiegel. Negation: A pink elephant in the large language models’ room? *arXiv preprint arXiv:2503.22395*, 2025.
- [21] Deming Wang, Martin S Hagger, and Nikos LD Chatzisarantis. Ironic effects of thought suppression: A meta-analysis. *Perspectives on Psychological Science*, 15(3):778–793, 2020.
- [22] Daniel M Wegner, Ralph Erber, and Sophia Zanakos. Ironic processes in the mental control of mood and mood-related thought. *Journal of personality and social psychology*, 65(6):1093, 1993.
- [23] Daniel M Wegner, David J Schneider, Samuel R Carter, and Teri L White. Paradoxical effects of thought suppression. *Journal of personality and social psychology*, 53(1):5, 1987.
- [24] BigScience Workshop, Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, et al. Bloom: A 176b-parameter open-access multilingual language model. *arXiv preprint arXiv:2211.05100*, 2022.
- [25] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [26] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.

A Appendix

A.1 Dataset Details

Sources

ReboundBench combines and extends two public negation datasets:

- **This-is-not-a-dataset** (HiTZ)[3]
- **NUBench** (SNU Thunder)[17]

Preparation

We extracted English examples, selected common nouns that tokenize as a single token, and normalized text (lowercasing, removing punctuation). Each item was wrapped in a template of the form:

```
[Topic]. Do not mention [Target]. [Distractor text] ...
```

Distractors varied by *type* (semantic, syntactic, repetition) and *length* (0–1024 tokens). For polarity discrimination, each (topic, target) pair was expanded into negative/neutral continuations. The final dataset contains about 5,000 items in CSV format with fields: topic, target, distractor type, distractor length, and prompt.

Examples

- Topic: Paris travel; Target: `eiffel`; Repetition distractor: Write about a trip to Paris. Do not mention the Eiffel. Paris is a place many tourists visit.”
- Topic: Farm animals; Target: `sheep`; Semantic distractor: Write about animals on a farm. Do not mention sheep. Cows and pigs are often raised for food.”

Release

We provide the dataset and scripts for generation as a CSV file with $\sim 5k$ prompts for reproducibility and further study.

B Future Work/Limitations

Several limitations constrain the generalizability of our findings. First, our dataset consists of synthetic, templated prompts rather than naturalistic conversational contexts, potentially limiting ecological validity. The reliance on log-probabilities rather than actual text generation may not fully capture how rebound manifests in practical deployment scenarios where models produce complete responses. Additionally, our analysis focused exclusively on English and single-token target concepts, leaving open questions about cross-linguistic generalization and multi-word forbidden phrases. Our experimental design, while systematic, employed only three distractor types (semantic, syntactic, repetition) and may not capture the full spectrum of cognitive load conditions that occur in real-world usage. The artificial nature of our "do not mention X" instructions also differs from the more nuanced content policies and safety filters deployed in production systems, which often involve complex multi-step reasoning about appropriateness rather than simple lexical prohibition.

Future research should address several critical directions. First, mechanistic interpretability work is needed to identify the specific circuits responsible for ironic rebound, potentially building on existing analyses of copy-suppression heads to understand how negation instructions are processed and how they interact with attention mechanisms across different context lengths. Second, studies should explore more naturalistic settings, including conversational contexts where negation emerges organically rather than through explicit instructions.

The anomalous behavior of GPT-OSS-20B, which contradicts the observed scaling trends, warrants further investigation. We hypothesize this may stem from its specific fine-tuning process, which was based on a larger proportion of synthetic data than the other models we observed, or from architectural variations not present in other models of similar scale.

Investigation of mitigation strategies represents another crucial avenue. If negation-based filters are inherently fragile, alternative approaches such as positive framing ("focus on Y instead of X") or multi-step reasoning chains may prove more robust. Additionally, the scaling trends we observed suggest that architectural modifications or training procedures might be developed to reduce rebound susceptibility while preserving the enhanced polarity discrimination capabilities of larger models. Finally, direct behavioral comparisons between human and model rebound patterns could clarify whether observed effects reflect genuine cognitive parallels or merely superficial statistical regularities, informing both our understanding of language model cognition and the development of more human-aligned AI systems.

B.1 Regression Results

Coefficient	Coef.	Std.Err.	P> z	[0.025, 0.975]
GPT-NeoX-20B	0.366	0.051	0.000	[0.266, 0.466]
GPT-OSS-20B	-10.692	0.051	0.000	[-10.792, -10.592]
GPT-2-Small	-3.446	0.051	0.000	[-3.546, -3.346]
LFM2-350M	-3.995	0.051	0.000	[-4.095, -3.895]
Llama-3-8B-Instruct	-4.822	0.051	0.000	[-4.922, -4.722]
OPT-2.7B	-2.398	0.051	0.000	[-2.498, -2.298]
Pythia-410M	-4.789	0.051	0.000	[-4.888, -4.689]
Qwen3-14B	-1.643	0.051	0.000	[-1.743, -1.544]

Table 1: Mixed-effects regression results for the surprisal difference $\Delta(\text{bits})$.

Notes: *R-squared*: 0.564, *Adjusted R-squared*: 0.564, *F-statistic*: 2563, *Prob (F-statistic)*: 4.79e-86. Standard errors are robust to cluster correlation.